Low-Dose CT Image Reconstruction Method With Probabilistic Atlas Prior

Mona Selim, Hiroyuki Kudo and Essam A. Rashed

***Abstract*—This work investigates the problem of image reconstruction from low-dose x-ray computed tomography (CT). Statistical iterative reconstruction is known to provide higher image quality due to the ability to incorporate prior knowledge to the reconstruction method and accurately model the photon statistics. In this paper, we develop a statistical reconstruction method using prior knowledge extracted from probabilistic atlas. First, we use a set of CT images previously scanned of various patients to generate a probabilistic atlas using Gaussian mixture model (GMM). Then, expectation maximization (EM) clustering algorithm is used to estimate the mixture parameters. Probabilistic atlas and mixture model parameters are then used to formulate the image reconstruction cost function. By merging the atlas information and smoothing penalty into the reconstruction procedure, image quality has been remarkably improved.**

I. Introduction

Computed tomography (CT) is considered one of the leading imaging modality in diagnostic imaging worldwide. Nevertheless, high radiation dose generated from CT scanning in several clinical applications consider a potential risk to the patients. It is not yet strictly proved that regular CT scans may cause a malignancy. However, it is estimated that a rough of 2% of cancers may eventually be caused by the average radiation dose currently used in clinical CT [1]. It is known that adjusting the x-ray tube voltage (kVp), tube current (mAs), pitch value and number of projection views around the body is an effective factors to execute radiation dose reduction. However, the resultant image usually suffers from serious noise and artifacts by using analytical reconstruction methods. It is always preferable to have standard imaging techniques that diminish the patient dose with reasonable image quality. In this paper, we aim to reduce high radiation dose and we depend on lowering x-ray tube current to achieve this aim.

Several image reconstruction methods have been developed to deal with low-dose CT imaging, for example, see [2]–[7]. Statistical iterative reconstruction (SIR) is known to provide higher image quality due to the ability to incorporate prior knowledge to the reconstruction method and accurately model the photon statistics. Commonly, the SIR techniques can be derived from the maximum *a posteriori* (MAP) estimator given the observed data or measurement which usually compose of two parts in the associative cost function. Definitely, the first part called the data-fidelity term and the second part called regularization term or penalty term. The data-fidelity part is responsible for modeling the statistical measurements, and the regularization term considers *a priori* knowledge that penalizes the solution.

Numerous prior options have been investigated in the past decades. The possible prior information that participate in the image quality can be categorized into three groups as follow: (1) full reference image (intensity value + pixel position), (2) boundaries and support information (pixel position only) and (3) attenuation information (intensity value only). The first group reflect the situation where the priori knowledge is taken from a reference image produced prior to the reconstruction process. The reference image idiom always indicates to an image that involves normal and basic structures of the scanned image. The reference image may be acquired within an old scan of the patient, segmentation of neighboring slices or a blurred image reconstructed from all measured projection data in gated-CT. The second group depicts the use of a prior knowledge according to the boundaries of uniform regions without additional spatial information. This involves the well-known quadratic smoothing penalty. The third group reflects the situation where priori knowledge is defined to a set of intensity values that represent uniform regions of the image without further information about the pixel positions.

In this paper, we investigate a new SIR method using prior information formulated from probabilistic atlas. A set of reconstructed volumes of previous scans of patients are used to construct a probabilistic atlas. The image reconstruction cost function is designed to incorporate the probabilistic atlas information and smoothing penalty to the reconstruction procedure. Evaluating the proposed method by simulated and real data proved that the proposed method can improve the quality of the images.

The organization of this paper is as follow. In section II, we introduce briefly the technique used for the construction of the probabilistic atlas. Then, the proposed image reconstruction algorithm is detailed. In section III, we present and analyze the experimental results that include simulation and real data. Finally, conclusion is showed in section IV.

II. Proposed Method

A. Construction of the probabilistic atlas

The probabilistic atlas is constructed as follow. First, the image dataset is co-registered using intensity-based automatic technique [8]. Definitely, we used mean square error as similarity metric and regular step gradient descent as optimizer.

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M.Selim is with the Dept. of Mathematics and Computer Science, Faculty of Science, Suez University, Suez, Egypt (E-mail: mona.selim@suezuniv.edu.eg). H.Kudo is with the Division of Information Engineering, Faculty of Engineering, Information and Systems, University of Tsukuba, Tsukuba, Japan and E.A.Rashed is with the Dept. of Mathematics, Faculty Science, Suez Canal University, Ismailia 41522, Egypt.

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Also, the transformation model used is affine which depicted by (translations, rotations, scalings and shears) parameters. Second, the EM algorithm [10] is used to segment the dataset into L clusters based on GMM. Each cluster consists of image pixels that represents a single region (organ) of the patient. The probabilistic density function (pdf) of GMM is denoted by

$$p(x_j) = \sum_{l=1}^{L} \pi_l \; p(x_j|\Omega_l), \quad j = 1, \ldots, n \tag{1}$$

$$0 \leq \pi_l \leq 1 \quad and \quad \sum_{l=1}^{L} \pi_l = 1, \tag{2}$$

$$p(x_j|\Omega_l) = \frac{1}{\sqrt{2\pi_l \sigma_l^2}} \exp\left(-\frac{(x_j - \mu_l)^2}{2\sigma_l^2}\right), \tag{3}$$

where $p(x_j|\Omega_l)$ represent a component of the mixture with mean μ_l and covariance σ_l, $\vec{x} = (x_1, \ldots, x_n)$ is the image vector, Ω_l $(l = 1, \ldots, L)$ is the class labels and π_l is the prior probability for each pixel x_j to belong to the class Ω_l and satisfies the constraints in Eq. (2). The well-known EM algorithm is used to estimate GMM parameter (π_l, μ_l, σ_l) and thus update the posterior probability which is denoted by

$$p(\Omega_l|x_j) = \frac{\pi_l p(x_j|\Omega_l)}{\sum_{l=1}^{L} \pi_l p(x_j|\Omega_l)}. \tag{4}$$

Given the posterior probability, a reference image $\vec{u} = (u_1, \ldots, u_j)$ is computed as the class mean value with higher probability component as follow

$$u_j = \{\mu_{\hat{l}} \mid p(\Omega_{\hat{l}}|x_j) = \max_{l=1}^{L} \; p(\Omega_l|x_j)\}. \tag{5}$$

Figure 1 illustrate an example of reference image $\vec{u}$ generation from probabilistic atlas of three clusters using Eq. (5). Each pixel value in the reference image is computed as the mean value μ_l of the cluster l.

B. Image reconstruction framework

The x-ray CT measurements can be formulated in the following statistical framework

$$y_i \approx \text{Poisson}\; [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle)] \tag{6}$$

where $\vec{y} = (y_1, \ldots, y_m)$ is the detector measurements with blank scan $\vec{b} = (b_1, \ldots, b_m)$ and $A = \{a_{ij}\}$ represents $m \times n$ system matrix. The proposed image reconstruction method is derived by minimizing the following cost function:

$$f(\vec{x}) = L(\vec{x}) + \vec{\gamma} D(\vec{x}, \vec{u}) + \beta R(\vec{x}), \tag{7}$$

where $L(\vec{x})$ is the negative log-likelihood function given by

$$L(\vec{x}) = \sum_{i=1}^{m} [b_i \exp(-\langle \vec{a_i}, \vec{x} \rangle) + y_i \langle \vec{a_i}, \vec{x} \rangle], \tag{8}$$

and $D(\vec{x}, \vec{u})$ is a distance function between the target image $\vec{x}$ and the reference image $\vec{u}$. The parameter $\vec{\gamma} = (\gamma_1, \ldots, \gamma_n)$ is the atlas weight where γ_j is computed as follow

$$\gamma_j = \max_{l=1}^{L} \; p(\Omega_l|x_j) \tag{9}$$

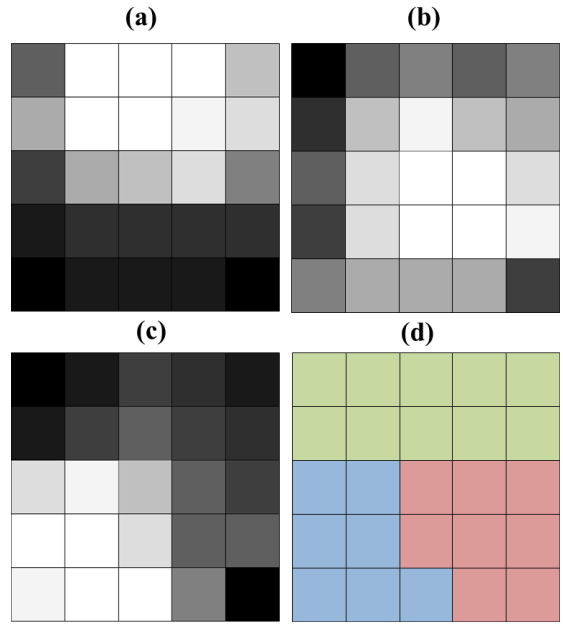

Fig. 1. Example of reference image generation from probabilistic atlas of three clusters. (a)-(c) are the probability components associated to clusters 1-3, respectively. White and black colors define the probabilities of one and zero, respectively. (d) is the reference image generated using Eq. (5) where green, red and blue are define pixels belongs to clusters 1, 2 and 3, respectively.

Here, we define $D(.,.)$ as the ℓ_1-norm distance and is given by

$$D(\vec{x}) = \sum_{j=1}^{n} d(x_j, u_j), \qquad d(a, b) = |a - b|. \tag{10}$$

The function $R(\vec{x})$ is a smoothing penalty given by

$$R(\vec{x}) = \sum_{j} \sum_{\tilde{j} \in N_j} w_{j\tilde{j}} \psi(x_j - x_{\tilde{j}}), \; \psi(t) = \log\left(1 + \frac{|t|}{\delta}\right). \tag{11}$$

where N_j is the set of eight neighbors of j-th pixel and $w_{j\tilde{j}}$ is weighting parameter equals to 1 in vertical and horizontal neighboring pixels, and equals to $(1/\sqrt{2})$ for diagonal directions. Both penalty terms are controlled using parameters $\vec{\gamma}$ and β. The atlas prior parameter $\vec{\gamma}$ is computed as a pixel-dependent parameter and is computed as the maximum posterior probability associated with image pixel x_j as defined in Eq. (9). This means the effect of atlas prior is increased when the atlas prior information is of high probability (e.g. for pixels located inside distinct regions). On the other hand, the effect of atlas prior is weaken when there is high uncertainty of atlas information (e.g. for pixels located around boundaries between distinct regions).

The cost function in Eq. (7) is designed to look for the suitable solution that fit with the data fidelity term and reduce the effect of statistical noise using information obtained from the atlas. During the implementation of the method, we have observed that image pixels located close the region boundaries are still suffering from statistical noise. This is due to the weakness of atlas prior in these regions where the degree of uncertainty is relatively high. This problem can be solved by including a weak smoothing prior to the

const function. The cost function in Eq. (7) is approximated using the majorization−minimization technique similar to the study in [9]. The iterative reconstruction algorithm can be summarized as:

1 Initialization $x^{(0)} = \epsilon$, with $\epsilon > 0$, $k \leftarrow 0$
repeat
  for $j = 1 : n$ **do**
2 $d_j = -\beta \sum_{\tilde{j} \in N_j} w_{j\tilde{j}} (x_j - x_{\tilde{j}}) \dot{\psi}(x_j - x_{\tilde{j}})$
3 $s_j = 2\beta \sum_{\tilde{j} \in N_j} w_{j\tilde{j}} x_j \dot{\psi}^2 (x_j - x_{\tilde{j}})$
4 $p_j = x_j^k + x_j^k \frac{d_j + \sum_i a_{ij} \left[b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle) - y_i \right]}{s_j + \sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle)}$
5 $t_j = \gamma_j x_j^k / (s_j + \sum_i a_{ij} \langle \vec{a_i}, \vec{x}^k \rangle b_i \exp(-\langle \vec{a_i}, \vec{x}^k \rangle))$
6 $x_j^{k+1} = Th(p_j, u_j, t_j)$
  end
  $k \leftarrow k + 1$
until *stopping criterion*;

where the thresholding function in step (6) is given by

$$Th(a,b,c) = \begin{cases} a - c & (a > b + c) \\ a + c & (a < b - c) \\ b & (otherwise) \end{cases} . \tag{12}$$

III. Experimental results

The proposed method is evaluated in this section. The experiments are executed using a simulation data and real data.

A. Simulated data

We carried out a simulation study to evaluate the proposed method. A set of 30 digital phantoms composed of five clusters are used to construct the atlas. Simulated organs are set to have small variation in size. The image size is 256×256 pixels with pixel size of 1.25×1.25 mm. The number of iterations k is set to 30, the number of detected photons are 5×10^4 photons, and the number of projections is 256. To generate the atlas, we selected one image as reference and the other 29 images are registered onto the reference image using intensity based registration with affine transformation and mean square error similarity metric. The EM algorithm is used to estimate the GMM parameters (π_l, μ_l, σ_l) and thus the posterior probability function. Figure 2 illustrate the atlas of simulation data. Also, we evaluated the proposed method by digital phantom like the phantoms used in atlas construction with small lesion insert. We have implemented the standard filtered back-projection (FBP) and the proposed method. From the results shown in Fig. 3, it can be deduced that the proposed method achieves better results even with small lesion that does not exist in the images used in atlas construction. Generally, it can be observed that using the probabilistic atlas in image reconstruction enhance the quality of the image.

B. Real data

A dataset composed of 68 screening chest CT volumes of various patients each of 25 to 31 slices with size 320×320 pixels and pixel size of 1×1 mm. A subset of 55 volumes were chosen to construct the atlas. A single volume is selected as a reference and the residual 54 volumes are registered to the reference volume and the atlas is constructed by the same strategy used in simulation data. The atlas is modeled to consist of six clusters (bone, muscles, fatty-tissues, lungs, air, bed) as shown in Fig. 4.

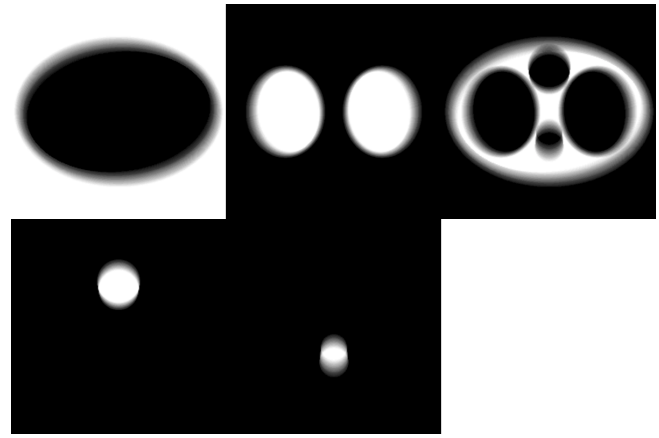

Fig. 2. Probabilistic atlas ($L = 5$) created from 30 digital phantoms with different sizes and angles of their mixtures using Gaussian mixture model (white is 1 and black is 0 values).

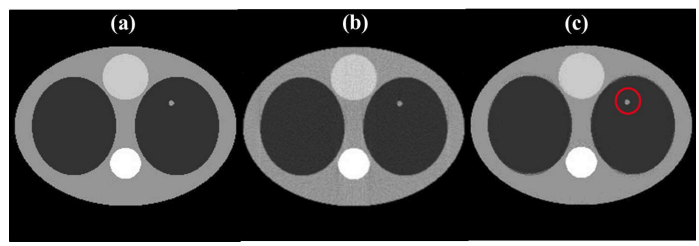

Fig. 3. (a) True object, (b) FBP reconstruction and (c) image reconstructed using the proposed method.. Display gray scale is (0.0, 0.18) cm^{-1}.

To evaluate the proposed method, we implemented the FBP algorithm and the proposed method on central slice of single volume which is not included in the construction process of the atlas. The number of detected photons per detector channel is 5×10^4 photons, the number of iterations is 50, and the number of projection views is 256. Figures 5 and 6 illustrate central reconstructed slices using FBP and proposed method of two different patients. The FBP image suffers from noise which is effectively decreased by using the proposed methods. We have used root mean square error (RMSE) as evaluation metric of image quality and results are shown in Fig. 7. It is clear that using the probabilistic atlas in image reconstruction give better image quality.

IV. Conclusion

In this paper, we investigate a statistical reconstruction method to reduce the radiation dose in x-ray CT. We incorporated smoothing penalty and prior information from probabilistic atlas to the image reconstruction cost function. We created the probabilistic atlas by using image registration and segmentation techniques. We evaluated the proposed method by using simulated and real data. In real data, chest screening CT images are used in creating atlas and testing the proposed

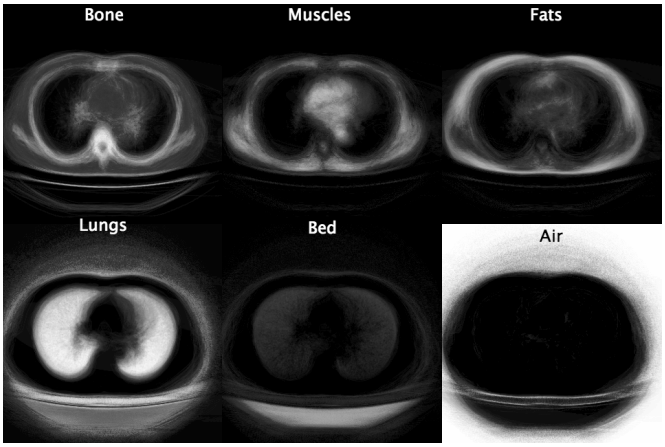

Fig. 4. Probabilistic atlas ($L = 6$) created from chest screening CT images of 55 patients using Gaussian mixture model (white is 1 and black is 0 values).

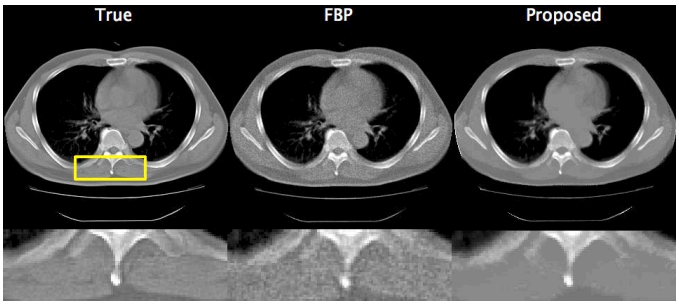

Fig. 5. Central slice of patient number 2 of the chest CT dataset. The first row shows original slice image and reconstructions using FBP, and the proposed method. Below is a zoomed display of regions marked with yellow rectangle. Display gray scale is (0.4, 1.6) cm^{-1}.

method. The experimental results showed that the proposed method provide higher image quality.

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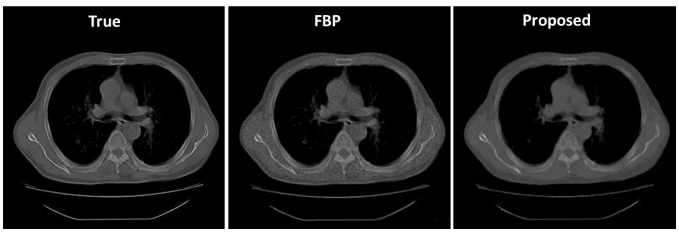

Fig. 6. Central slice of patient number 43 of the chest CT dataset. The first row shows original slice image and reconstructions using FBP, and the proposed method. Display gray scale is (0.4, 1.6) cm^{-1}.

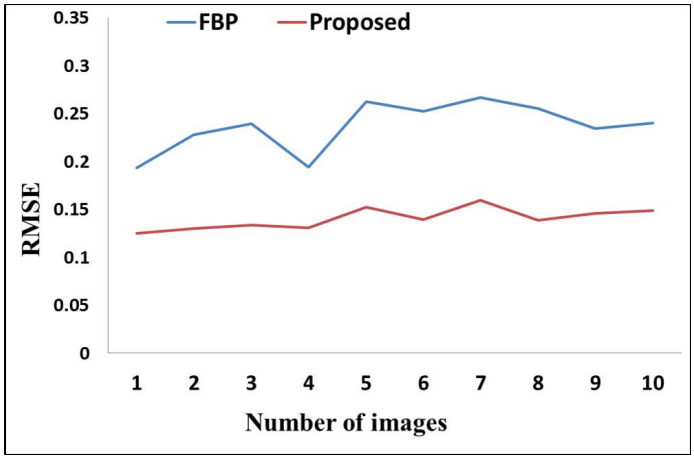

Fig. 7. The value of Root Mean Square Error (RMSE) for proposed method and FBP method for 10 images.

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